SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Project Initialization and Planning Phase

Date	15 March 2026
Team ID	xxxxxx
Project Name	**OptiCrop: Smart Agricultural Production Optimization Engine**
Maximum Marks	3 Marks

Define Problem Statements (Customer Problem Statement Template):

I am	I'm trying to	But	Because	Which makes me feel
OptiCrop – Smart Agricultural Production Optimization Engine an intelligent system that helps farmers make data-driven decisions to improve crop productivity and resource efficiency.	use machine learning and data analytics to analyze soil, weather, crop, and environmental factors to recommend the best crops and optimize agricultural practices for higher yield and sustainability.	traditional methods rely on manual analysis and experience, which is time-consuming, inconsistent, and may not always produce the most effective results.	there is no automated system that comprehensively analyzes multi-source agricultural data and provides accurate, real-time optimization recommendations.	confident that OptiCrop can empower farmers with smart insights, reduce risk, save resources, and maximize agricultural productivity in a sustainable way.

Project: OptiCrop – Smart Agricultural Production Optimization Engine using Machine Learning

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1 Farmer	a small-scale farmer who wants to get the best crop recommendations for my field	choose the most suitable crops based on soil, weather, and season to maximize yield	I manually analyze information from different sources, which takes time and may not always be accurate	there is no automated system that analyzes multiple agricultural factors and provides reliable crop recommendations	uncertain and worried about low productivity and financial losses
PS-2 Agricultural Advisor	an agricultural advisor who supports farmers with better farming decisions	provide data-driven recommendations and monitor crop performance for farmers	traditional methods lack real-time insights and accurate predictions across different regions and conditions	the data is scattered across multiple sources and there is no unified platform for analysis and optimization	concerned about giving inconsistent advice and losing farmers' trust
PS-3 Researcher / Data Analyst	a researcher who analyzes agricultural data to improve farming strategies and sustainability	build and evaluate machine learning models that predict crop suitability and optimize resources	I lack an integrated system with quality data, visualization, and evaluation tools to validate model performance	there is no end-to-end platform that combines machine learning, model evaluation, and data visualization	frustrated and limited in creating impactful and scalable solutions